The Gamma Function

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∞ Let Infinity Be Your Limit ∞

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Introduction

We all know about Factorials. That's a quite interesting notion- Multiplying the integer with all the previous integer till 1 *e.g.*

$$67! = 67 \times 68 \times \cdots \times 3 \times 2 \times 1$$

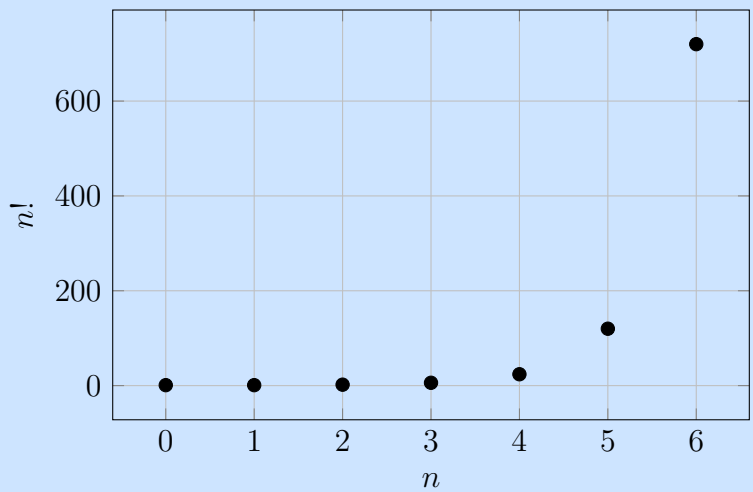
But those were only about positive integers, namely, Natural Numbers. What about the other numbers that exists? I am talking about *ANY* number- $1, 1.2, \frac{1}{2}, 67.67$ even irrationals- π, π^e

Let's proceed our discussion with plotting some points for $n!$ with positive integer values of n-

Plotting values of $n!$ for $n \in \mathbb{N}$

n	**n!**
0	1
1	1
2	2
3	6
4	24
5	120
6	720

Scatter Plot of n vs $n!$ (for $0 \leq n \leq 6$)



Notice that, the value of $n!$ is growing exponentially. Now the core part- to get the value of $n!$ for *any* n between the black dots (*i.e.* for integer $n!$), we want an *appropriate* function, whose graph connects with dots. Here's what I meant-

Finding appropriate "curves" to fit in

Take the point $n = 5.5$ in the graph for example (denoted with red dot), we don't know where the y value (i.e. $y = n!$) will belong, so we want a "curve" that APPROPRIATELY connects with the points.

When I am using the word **APPROPRIATELY**, i am referring to something rather important- there can be lot of "curves" corresponding to some functions that "connects" with the points but have absolutely NO relation with the function $y = n!$. Here follows some examples-

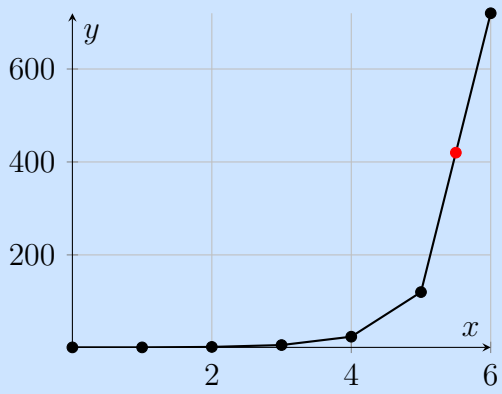


Diagram-1

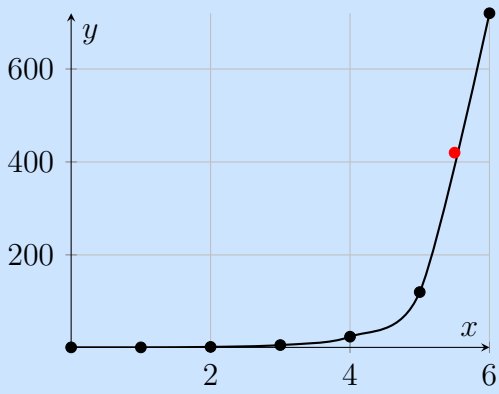


Diagram-2

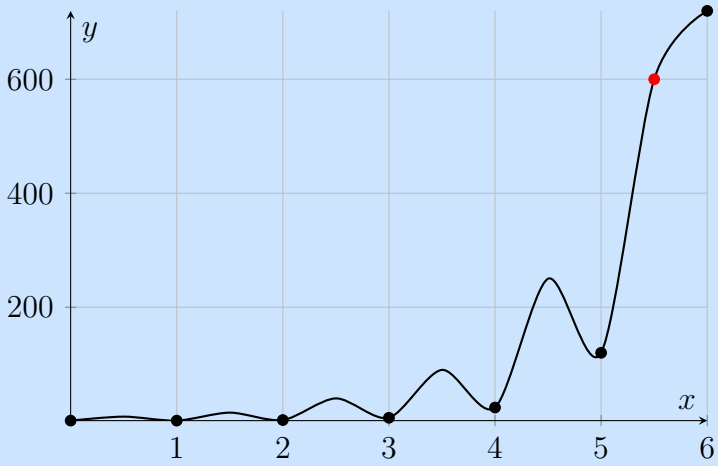


Diagram-3

Notice that, based on the specific curves (the first one isn't a curve though), the value of 5.5! is varying. So we need a function $f(n)$ that is equal to $n!$, within a domain in $\mathbb{R}$ and the corresponding graph (the "curve") will be our *APPROPRIATE** curve that perfectly connects with the dots and have direct relation with $n!$ and is with which we can factorial value for any real number. So what we want is, some $f(n)$ such that,

$$f(n) = n!$$

The Gamma Function

Apparently, mathematician Leonard Euler $(1707 - 1783)$ was having similar thoughts as we are having right now. Fortunately enough, while experimenting with numbers, he found an expression that perfectly matches with $n!$ -

$$\left[\left(\frac{1}{2}\right)^n \frac{1}{n+1}\right]\left[\left(\frac{2}{3}\right)^n \frac{2}{n+2}\right]\left[\left(\frac{3}{4}\right)^n \frac{3}{n+3}\right]\cdots = n!$$

The expression leonard found is now written conventionally with the use of the concept of Limits as,

$$\lim_{m\to\infty} \frac{m!\, m^{n+1}}{(n+1)(n+2)\cdots(m+n+1)}$$

However, Euler didn't stopped there, he went on and realised that the his finding of the new way of writing $n!$ is linked with an integral,

$$\int_0^1 x^m(1-x)^n\, dx = \frac{1\cdot 2\ldots n}{(m+1)(m+2)\ldots(m+n+1)}$$

After some clever but cumbersome approaches, Euler finally derived the following integral in his original paper,

$$n! = \int_0^1 (-\log x)^n\, dx$$

Later on, Adrien Marie Legendre $(1752 - 1833)$ modified this integral to the following one,

$$\Gamma(z) = \int_0^\infty e^{-t} t^{z-1} /, dt$$

he also initiated the use of the Greek letter Γ "Gamma" . He mentioned the first integral $\int_0^1 x^m(1-x)^n\, dx$ to be First Euler Integral or "Euler's integral of the first kind" and the latter one as Second Euler Integral. The first Integral is now known as the Beta Function,

$$B(m,n) = \int_0^1 x^{m-1}(1-x)^{n-1}\, dx$$

I will not go in details regarding the Beta Function, as it demands a whole discussion about it.

Definition - Infinite Product

We define the define Gamma Function, denoted by $\Gamma(z)$, as follows

$$\Gamma(z) = \lim_{m\to\infty} \frac{m!\, m^{z+1}}{z(z+1)(z+2)\cdots(m+z)}$$

Where $z \in \mathbb{C}$ and $z \notin \{0, -1, 2, \dots\}$
And thus for the value $z+1$,

$$\Gamma(z+1) = \lim_{m\to\infty} \frac{m!\, m^{z+1}}{(z+1)(z+2)\cdots(m+z+1)}$$

Definition - with Integral

We define the define Gamma Function with integral as follows

$$\Gamma(z) = \int_0^\infty e^{-t} t^{z-1} /, dt$$

Where $z \in \mathbb{C}$ and $z \notin \{0, -1, 2, \dots\}$
And thus for the value $z+1$,

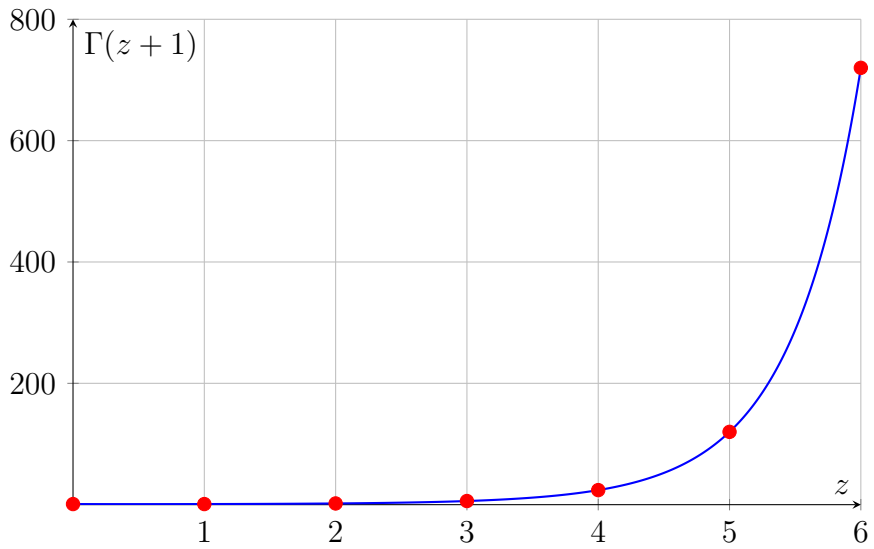
$$\Gamma(z+1) = \int_0^\infty e^{-t} t^{z} /, dt$$

Notice that in the definitions above, we have omitted using n and instead used z as the variable. The reason is the Gamma Function extends the definition of factorials beyond Real values to Imaginary Values. But a crucial thing is- We also dropped the negative integers from the definition. The reason is, the integral doesn't converge and overall behaviours goes nasty.

Also, note that we have the following property-

$$z! = \Gamma(z+1)$$

Now finally we have that **APPROPRIATE** relation for the curve that we discussed at the beginning and here it is-



You can get the value of any factorial value with gamma function, with the following python code,

Python code to get gamma values

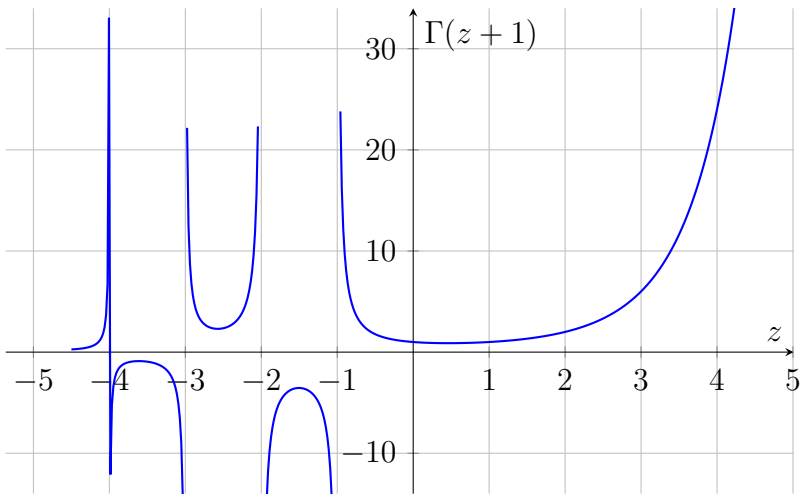
With this following code you can get any real number's factorial-

```
import numpy as np
from scipy.special import gamma
import math

x = 5.5 # Put your values here
y = gamma(x+1)

print(y)
```

make sure to intall scypy and numpy using pip-

```
pip install scypy numpy
```

Below is the graph of Gamma function including negative domain values. Notice $\Gamma(z)$ don't give any value for negative integers, but they correspond perfectly for Non-Integers-



1 Problems

P0

Prove that $\Gamma(z+1) = z\Gamma(z)$

P1 (Weierstrass Product)

Prove the following equation from the Infinite Product definition,

$$\frac{1}{\Gamma(z)} = z\, e^{\gamma z} \prod_{n=1}^{\infty} \left(1 + \frac{z}{n}\right) e^{-z/n}$$

where

$$\gamma = \lim_{n\to\infty} \left(\sum_{k=1}^{n} \frac{1}{k} - \ln n \right) \approx 0.5772156649$$

P2

Derive Euler's Second Integral from the First one

P3

Derive the integral definition from Euler's Second Integral

P4

Determine the value of

$$\Gamma(\frac{1}{2})$$

as a closed form relation.
(Spoiler: It's $\sqrt{\pi}$)